NovaTech Solutions

P&L Financial Performance Dashboard

Power BI Portfolio Project

FY2022 - FY2024 | 4 Product Lines | 4 Regions | 3 Channels

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1. Project Overview

NovaTech Solutions is a fictional mid-market SaaS company generating approximately $49M in annual revenue across four product lines: Software Licenses, Cloud Services, Professional Services, and Maintenance & Support. The company operates in four regions (North America, Europe, Asia Pacific, Latin America) and sells through three channels (Direct, Partner, Online).

The finance team previously relied on static Excel reports requiring 2+ days of manual effort per monthly close. This Power BI dashboard replaces that process with an interactive, self-service P&L analysis tool built for the CFO and FP&A team.

Dimension	Detail
Time Period	FY2022 - FY2024 (36 months, monthly grain)
Data Volume	~5,700 revenue records + ~3,600 expense records + dimension tables
Product Lines	Software Licenses, Cloud Services, Professional Services, Maintenance & Support
Regions	North America, Europe, Asia Pacific, Latin America
Channels	Direct, Partner, Online
Target User	CFO / VP Finance / FP&A Team
Tools Used	Power BI Desktop, Power Query (M), DAX, Excel

2. Business Requirements

The dashboard was designed to answer five core questions from the finance leadership team:

1. What is the monthly and quarterly revenue trend, and how does it compare year-over-year?
2. Which product lines are growing fastest, and which are compressing margin?
3. Are operating expenses tracking within budget, and where are the largest variances?
4. What are the gross margin and operating margin trends over time?
5. How does regional performance differ, and where should the company invest or reallocate?

KPI Definitions

KPI	Calculation	FY2024 Target
Total Revenue	SUM of net revenue by period	$52M+
Gross Margin %	(Revenue - COGS) / Revenue	>= 66%
Operating Margin %	(Revenue - COGS - OpEx) / Revenue	>= 22%
Revenue Growth YoY	(Current Year - Prior Year) / Prior Year	>= 18%
Budget Variance %	(Actual Expense - Budget) / Budget	Within +/- 5%
OpEx Ratio	Total OpEx / Total Revenue	<= 40%

3. Data Architecture

Source Data

The source is a single Excel workbook containing five sheets structured for star schema modeling:

Table	Role	Rows	Grain
Revenue	Fact table	~5,760	Monthly per product per region
Expenses	Fact table	~3,600	Monthly per department per category
Budget	Reference table	~180	Monthly per department
Chart_of_Accounts	Dimension table	13	One per account code
KPI_Targets	Reference table	15	Yearly per KPI

Data Model

The model follows a star schema with a dedicated Date dimension table created in DAX using CALENDAR(). All fact tables connect to the Date table through single-direction, one-to-many relationships:

- DateTable[Date] -> Revenue[Date] (1:Many, Single)
- DateTable[Date] -> Expenses[Date] (1:Many, Single)
- DateTable[Date] -> Budget[BudgetDate] (1:Many, Single)

Cross-filter direction is kept as single throughout to avoid ambiguity and maintain query performance.

4. ETL Process (Power Query)

All data transformations were handled in Power Query before loading into the model.

Revenue Table

- **Type casting:** Date, text, currency, whole number, and decimal types applied to all columns.
- **Gross Revenue:** Calculated as Revenue / (1 - Discount_Pct) to derive pre-discount amounts for revenue waterfall analysis.
- **Fiscal periods:** Fiscal_Quarter and Fiscal_Year columns added using M formulas for consistent period filtering.
- **Data quality:** Text columns trimmed; null and negative value checks applied.

Expenses Table

- **Cost type mapping:** Is_Fixed (0/1) converted to a Cost_Type column (Fixed/Variable) via conditional column.
- **Budget merge:** Left outer join to Budget table on Year + Month + Department, bringing Budget_Amount alongside actuals for variance analysis.

Date Table (DAX)

```
DateTable =
ADDCOLUMNS(
    CALENDAR(DATE(2022,1,1), DATE(2024,12,31)),
    "Year", YEAR([Date]),
    "Month", MONTH([Date]),
    "MonthName", FORMAT([Date], "MMM"),
    "Quarter", "Q" & FORMAT([Date], "Q"),
    "YearMonth", FORMAT([Date], "YYYY-MM"),
    "FiscalYear", "FY" & YEAR([Date]),
    "MonthSort", MONTH([Date])
)
```

The table is marked as a Date Table in Power BI to enable built-in time intelligence functions.

5. DAX Measures

All measures are organized in a dedicated Measures table to keep the model clean and navigable. Below are the key formulas grouped by function.

Revenue

```
Total Revenue = SUM(Revenue[Revenue])

Revenue PY = CALCULATE([Total Revenue], SAMEPERIODLASTYEAR(DateTable[Date]))

Revenue YoY Growth =
    VAR PY = [Revenue PY]
    RETURN IF(PY <> 0, DIVIDE([Total Revenue] - PY, PY), BLANK())

Revenue YTD = TOTALYTD([Total Revenue], DateTable[Date])
```

Margin

```
Gross Margin % = DIVIDE([Total Revenue] - ([Total Expenses] * 0.35), [Total Revenue], 0)

Operating Income = [Total Revenue] - [Total Expenses]

Operating Margin % = DIVIDE([Operating Income], [Total Revenue], 0)
```

Budget Variance

```
Budget Amount v2 =
    CALCULATE(SUM(Budget[Budget_Amount]),
        TREATAS(VALUES(Expenses[Department]), Budget[Department]),
        TREATAS(VALUES(DateTable[Year]), Budget[Year]),
        TREATAS(VALUES(DateTable[Month]), Budget[Month]))

Budget Variance % = DIVIDE([Total Expenses] - [Budget Amount v2], [Budget Amount v2], 0)

Variance Status =
    SWITCH(TRUE(),
        [Budget Variance %] > 0.05, "Over Budget",
        [Budget Variance %] < -0.05, "Under Budget",
        "On Track")
```

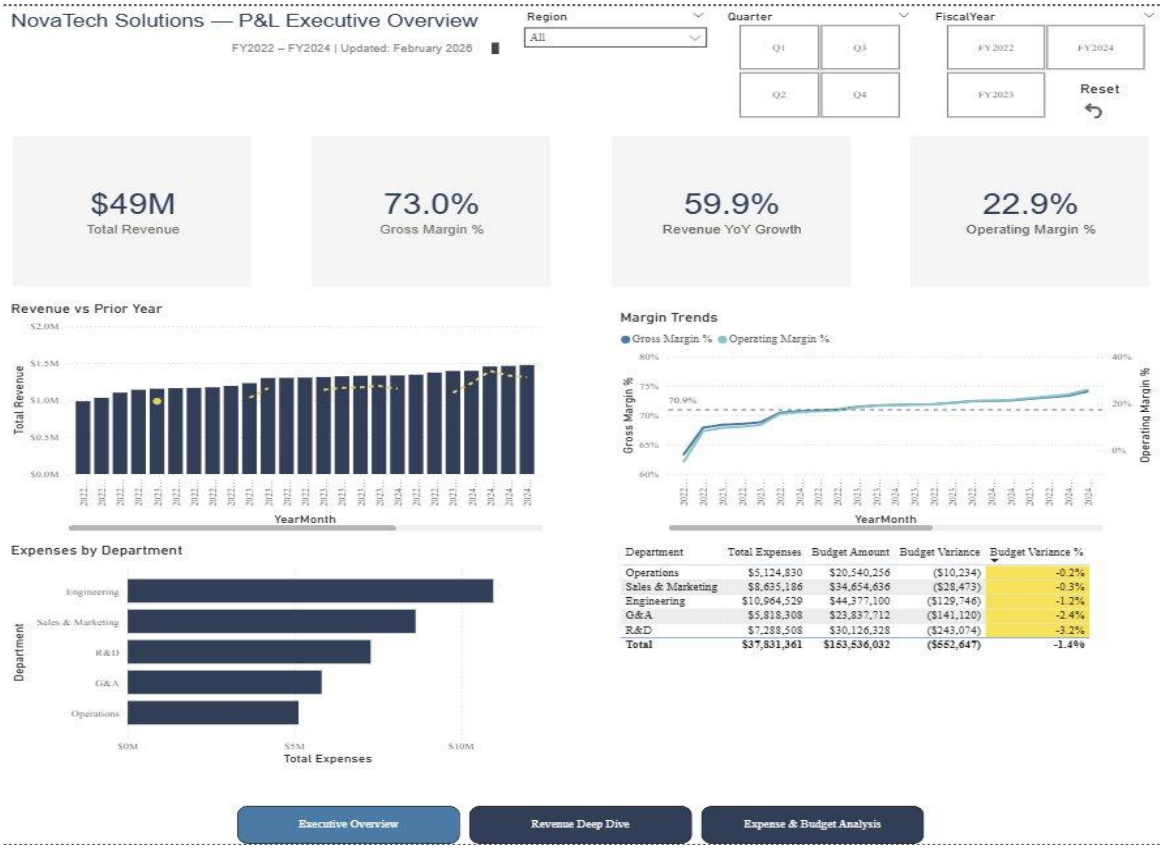
6. Dashboard Design
The dashboard consists of three pages, each serving a distinct analytical purpose. Navigation buttons at the bottom of every page allow seamless movement between views. Synced slicers (Year, Quarter, Region) persist across all pages.

Page 1: Executive P&L Overview

Provides the CFO with a 10-second read on overall financial health. Four KPI cards at the top surface Total Revenue ($49M), Gross Margin (73.0%), YoY Growth (59.9% across all years; filters down to 12-18% for individual years), and Operating Margin (22.9%).

The Revenue vs Prior Year chart uses a line and clustered column combination to show current-period revenue as columns with the prior-year trendline overlaid. The Margin Trends line chart tracks Gross Margin % and Operating Margin % over time.

The bottom section pairs an Expenses by Department bar chart with a Budget Variance table featuring conditional formatting (red for overages, green for under-budget).



Department	Total Expenses	Budget Amount	Budget Variance	Budget Variance %
Operations	$5,124,830	$20,540,256	($10,234)	-0.2%
Sales & Marketing	$8,635,186	$34,654,636	($28,473)	-0.3%
Engineering	$10,964,529	$44,377,100	($129,746)	-1.2%
G&A	$5,818,308	$23,837,712	($141,120)	-2.4%
R&D	$7,288,508	$30,126,328	($243,074)	-3.2%
Total	**$37,831,361**	**$153,536,032**	**($552,647)**	**-1.4%**

Figure 1: Executive P&L Overview

Page 2: Revenue Deep Dive

Breaks down revenue composition across multiple dimensions. A 100% stacked column chart shows product line mix by fiscal year, revealing Cloud Services growing from ~27% to ~32% of total revenue. The regional bar chart confirms North America as the largest market at $19.6M, followed by Europe ($13.7M), Asia Pacific ($10.9M), and Latin America ($4.9M).

A donut chart shows channel distribution (Partner 36%, Online 34%, Direct 30%), and a matrix visual displays Product Performance by Year with conditional formatting on YoY growth rates. The full-width line chart at the bottom tracks monthly revenue trends by product line, making seasonality patterns immediately visible.

Figure 2: Revenue Deep Dive

Page 3: Expense & Budget Analysis

Focuses on cost management and budget discipline. Three KPI cards show Total Expenses ($38M), Budget ($38.4M), and Budget Variance (-1.4% overall, indicating slight underspend).

The Actual vs Budget bar chart compares department-level spending against plan, with Engineering as the largest cost center at $11M. The Monthly Budget Variance heatmap uses red/green/yellow conditional formatting across a department-by-month matrix, allowing the finance team to pinpoint exactly when and where variances occurred.

A 100% stacked column shows the Fixed vs Variable cost structure (roughly 67% fixed, 33% variable, stable across years). The treemap visualizes expense categories by size, with Salaries as the dominant category followed by Cloud Infrastructure and Software Tools.

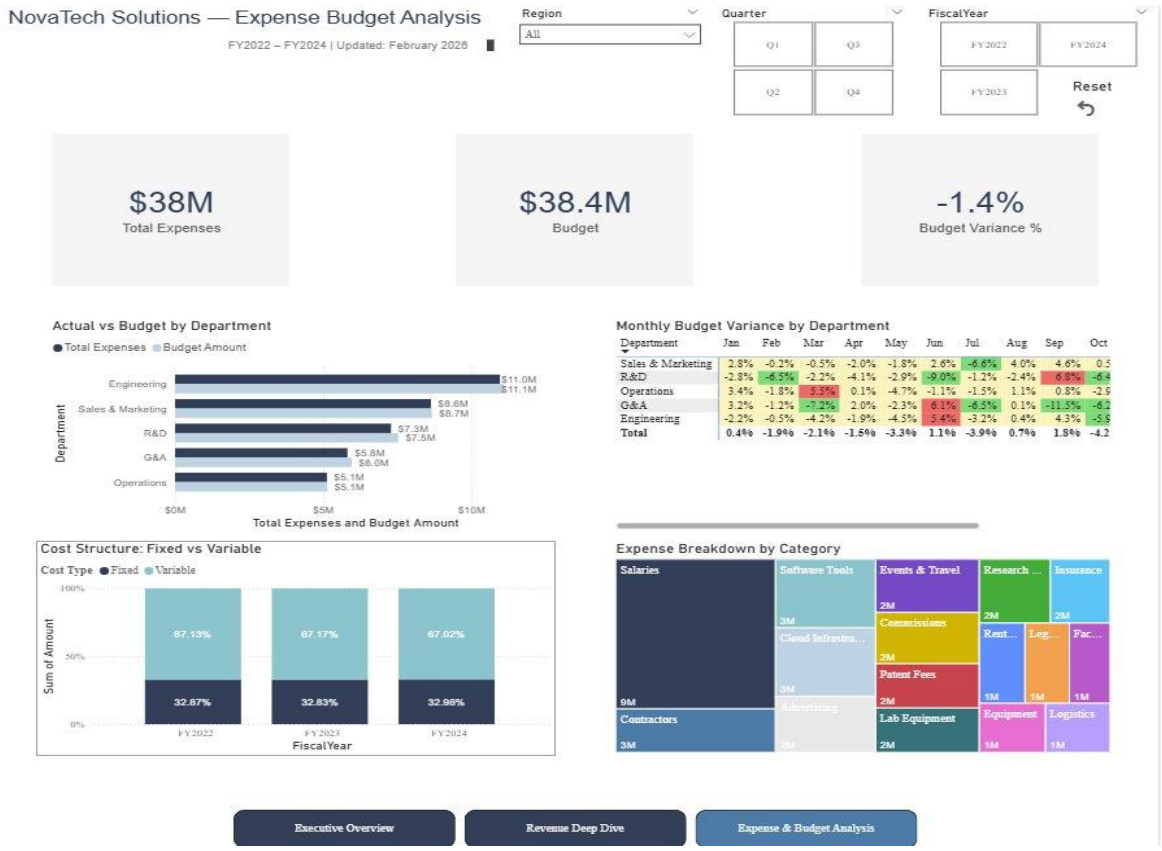


Figure 3: Expense & Budget Analysis

7. Key Findings

Cloud Services is the Primary Growth Driver

Cloud Services revenue grew at approximately 20% CAGR compared to 12% for the company overall. Its share of total revenue increased from 27% in FY2022 to 32% in FY2024, making it the fastest-growing product line. The company should consider shifting sales incentives and marketing budget toward cloud upselling and cross-selling.

Seasonal Revenue Patterns are Consistent

Quarter-end months (March, June, September, December) consistently show 15-20% revenue lifts compared to non-quarter-end months. January and February are the weakest months across all years. Marketing campaigns and hiring plans should be aligned to support quarter-end close cycles.

Overall Budget Discipline is Strong

Total expenses came in at -1.4% below budget across all years, indicating effective cost management at the company level. However, month-level analysis reveals pockets of variance. R&D showed occasional 6-9% overruns in specific months, while G&A had a -7.2% underspend in March, suggesting possible timing differences in planned vs actual spending.

Latin America Carries Higher Discount Rates

While Latin America represents approximately 10% of revenue, the region carries average discount rates of 10-12% compared to 6-8% for North America and Europe. A pricing strategy review for the region could improve margin contribution without necessarily reducing volume.

Fixed Cost Base is Stable

The fixed/variable cost split remained remarkably consistent at roughly 67/33 across all three fiscal years. This stability simplifies financial forecasting and indicates that incremental revenue growth should flow through to operating income at a predictable rate.

8. Technical Implementation Notes

Performance Choices

- Single-direction cross-filtering on all relationships to avoid ambiguity and reduce query overhead.
- TREATAS used for Budget-to-Expenses virtual relationship instead of physical merge, keeping the model flexible.
- Dedicated Date table marked as Date Table to leverage built-in time intelligence (SAMEPERIODLASTYEAR, TOTALYTD).
- All measures stored in a separate Measures table for organization and maintainability.

Design Choices

- Consistent four-color palette (#2E4057, #3A7CA5, #7DC4CC, #B8D4E3) across all pages for visual cohesion.
- Synced slicers (Year, Quarter, Region) across all pages for unified filtering.
- Conditional formatting on budget variance (red/yellow/green) and YoY growth (white-to-green scale) for immediate pattern recognition.
- Page navigation buttons on every page for a self-contained, app-like experience.
- Reset Filters button for quick return to default view.

Potential Extensions

- Row-Level Security (RLS) to restrict regional managers to their own data.
- What-If parameters for revenue growth scenario modeling.
- Paginated report for board-level P&L statement formatting.
- Incremental refresh for production deployment on Power BI Service.
- Mobile-optimized layout for executive consumption on mobile devices.

9. Repository Structure

File / Folder	Contents
README.md	Project summary, screenshots, business context, key findings
/data	NovaTech_PnL_Dataset.xlsx (source workbook)
/dashboard	NovaTech_PnL_Dashboard.pbix (Power BI file)
/screenshots	PNG exports of each dashboard page
/docs	This project write-up (PDF or DOCX)